

String :

1. Write a program to reverse a string.
2. Check if a string is a palindrome.
3. Convert a string to uppercase.
4. Convert a string to lowercase.
5. Count the number of vowels in a string.
6. Count the number of consonants in a string.
7. Remove all whitespaces from a string.
8. Find the length of a string without using the ``len()`` function.
9. Check if a string contains a specific word.
10. Replace a word in a string with another word.
11. Count the occurrences of a word in a string.
12. Find the first occurrence of a word in a string.
13. Find the last occurrence of a word in a string.
14. Split a string into a list of words.
15. Join a list of words into a string.
16. Convert a string where words are separated by spaces to one where words are separated by underscores.
17. Check if a string starts with a specific word or phrase.
18. Check if a string ends with a specific word or phrase.
19. Convert a string to title case (e.g., "hello world" to "Hello World").
20. Find the longest word in a string.
21. Find the shortest word in a string.
22. Reverse the order of words in a string.
23. Check if a string is alphanumeric.
24. Extract all digits from a string.
25. Extract all alphabets from a string.
26. Count the number of uppercase letters in a string.
27. Count the number of lowercase letters in a string.
28. Swap the case of each character in a string.
29. Remove a specific word from a string.
30. Check if a string is a valid email address.
31. Extract the username from an email address string.
32. Extract the domain name from an email address string.
33. Replace multiple spaces in a string with a single space.
34. Check if a string is a valid URL.
35. Extract the protocol (http or https) from a URL string.
36. Find the frequency of each character in a string.
37. Remove all punctuation from a string.
38. Check if a string contains only digits.
39. Check if a string contains only alphabets.
40. Convert a string to a list of characters.
41. Check if two strings are anagrams.

42. Encode a string using a Caesar cipher.
43. Decode a Caesar cipher encoded string.
44. Find the most frequent word in a string.
45. Find all unique words in a string.
46. Count the number of syllables in a string.
47. Check if a string contains any special characters.
48. Remove the nth word from a string.
49. Insert a word at the nth position in a string.
50. Convert a CSV string to a list of lists.

List:

1. Create a list with integers from 1 to 10.
2. Find the length of a list without using the `len()` function.
3. Append an element to the end of a list.
4. Insert an element at a specific index in a list.
5. Remove an element from a list by its value.
6. Remove an element from a list by its index.
7. Check if an element exists in a list.
8. Find the index of the first occurrence of an element in a list.
9. Count the occurrences of an element in a list.
10. Reverse the order of elements in a list.
11. Sort a list in ascending order.
12. Sort a list in descending order.
13. Create a list of even numbers from 1 to 20.
14. Create a list of odd numbers from 1 to 20.
15. Find the sum of all elements in a list.
16. Find the maximum value in a list.
17. Find the minimum value in a list.
18. Create a list of squares of numbers from 1 to 10.
19. Create a list of random numbers.
20. Remove duplicates from a list.
21. Find the common elements between two lists.
22. Find the difference between two lists.
23. Merge two lists.
24. Multiply all elements in a list by 2.
25. Filter out all even numbers from a list.
26. Convert a list of strings to a list of integers.
27. Convert a list of integers to a list of strings.
28. Flatten a nested list.
29. Create a list of the first 10 Fibonacci numbers.
30. Check if a list is sorted.

31. Rotate a list to the left by `n` positions.
32. Rotate a list to the right by `n` positions.
33. Create a list of prime numbers up to 50.
34. Split a list into chunks of size `n`.
35. Find the second largest number in a list.
36. Replace every element in a list with its square.
37. Convert a list to a dictionary where list elements become keys and their indices become values.
38. Shuffle the elements of a list randomly.
39. Create a list of the first 10 factorial numbers.
40. Check if two lists have at least one element in common.
41. Remove all elements from a list.
42. Replace negative numbers in a list with 0.
43. Convert a string into a list of words.
44. Convert a list of words into a string.
45. Create a list of the first `n` powers of 2.
46. Find the longest string in a list of strings.
47. Find the shortest string in a list of strings.
48. Create a list of the first `n` triangular numbers.
49. Check if a list contains another list as a subsequence.
50. Swap two elements in a list by their indices.

Tuple :

1. Create a tuple with integers from 1 to 5.
2. Access the third element of a tuple.
3. Find the length of a tuple without using the `len()` function.
4. Count the occurrences of an element in a tuple.
5. Find the index of the first occurrence of an element in a tuple.
6. Check if an element exists in a tuple.
7. Convert a tuple to a list.
8. Convert a list to a tuple.
9. Unpack the elements of a tuple into variables.
10. Create a tuple of even numbers from 1 to 10.
11. Create a tuple of odd numbers from 1 to 10.
12. Concatenate two tuples.
13. Repeat a tuple three times.
14. Check if a tuple is empty.
15. Create a nested tuple.
16. Access the first element of a nested tuple.
17. Create a tuple with a single element.

18. Compare two tuples.
19. Delete a tuple.
20. Slice a tuple.
21. Find the maximum value in a tuple.
22. Find the minimum value in a tuple.
23. Convert a string to a tuple of characters.
24. Convert a tuple of characters to a string.
25. Create a tuple from multiple data types.
26. Check if two tuples are identical.
27. Sort the elements of a tuple.
28. Convert a tuple of integers to a tuple of strings.
29. Convert a tuple of strings to a tuple of integers.
30. Merge two tuples.
31. Flatten a nested tuple.
32. Create a tuple of the first 5 prime numbers.
33. Check if a tuple is a palindrome.
34. Create a tuple of squares of numbers from 1 to 5.
35. Filter out all even numbers from a tuple.
36. Multiply all elements in a tuple by 2.
37. Create a tuple of random numbers.
38. Check if a tuple is sorted.
39. Rotate a tuple to the left by `n` positions.
40. Rotate a tuple to the right by `n` positions.
41. Create a tuple of the first 5 Fibonacci numbers.
42. Create a tuple from user input.
43. Swap two elements in a tuple.
44. Reverse the elements of a tuple.
45. Create a tuple of the first `n` powers of 2.
46. Find the longest string in a tuple of strings.
47. Find the shortest string in a tuple of strings.
48. Create a tuple of the first `n` triangular numbers.
49. Check if a tuple contains another tuple as a subsequence.
50. Create a tuple of alternating 1s and 0s of length `n`.

Set :

1. Create a set with integers from 1 to 5.
2. Add an element to a set.
3. Remove an element from a set.
4. Check if an element exists in a set.
5. Find the length of a set without using the `len()` function.
6. Clear all elements from a set.
7. Create a set of even numbers from 1 to 10.

8. Create a set of odd numbers from 1 to 10.
9. Find the union of two sets.
10. Find the intersection of two sets.
11. Find the difference between two sets.
12. Check if a set is a subset of another set.
13. Check if a set is a superset of another set.
14. Create a set from a list.
15. Convert a set to a list.
16. Remove a random element from a set.
17. Pop an element from a set.
18. Check if two sets have no elements in common.
19. Find the symmetric difference between two sets.
20. Update a set with elements from another set.
21. Create a set of the first 5 prime numbers.
22. Check if two sets are identical.
23. Create a frozen set.
24. Check if a set is disjoint with another set.
25. Create a set of squares of numbers from 1 to 5.
26. Filter out all even numbers from a set.
27. Multiply all elements in a set by 2.
28. Create a set of random numbers.
29. Check if a set is empty.
30. Create a nested set (hint: use frozenset).
31. Remove an element from a set using the discard method.
32. Compare two sets.
33. Create a set from a string.
34. Convert a set of strings to a set of integers.
35. Convert a set of integers to a set of strings.
36. Create a set from a tuple.
37. Convert a set to a tuple.
38. Find the maximum value in a set.
39. Find the minimum value in a set.
40. Create a set from user input.
41. Check if the intersection of two sets is empty.
42. Create a set of the first 5 Fibonacci numbers.
43. Remove duplicates from a list using sets.
44. Check if two sets have the same elements, regardless of their count.
45. Create a set of the first n powers of 2.
46. Find the common elements between a set and a list.
47. Create a set of the first n triangular numbers.
48. Check if a set contains another set as a subset.
49. Create a set of alternating 1s and 0s of length n .
50. Merge multiple sets into one.

